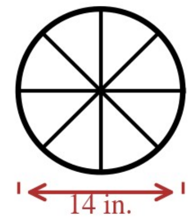
SC Honors Geometry - Nappier
11.2 Areas of Circles and Sectors

Speak **L**ike **A** **M**athematician:

sector of a circle:

the big idea:



Area Formulas

Area of a circle: A = π ______ .

Area of a sector: A = (arc measure / 360°) · ___________ .

Example 1 – Circle Areas

a) Find the area of a circle with radius 6 (exact and nearest tenth).

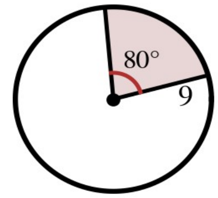
b) Find the area of a circle with diameter 20.

c) A circle has area 81π. Find its radius.

Example 2 – Sector Area

In the top figure, the sector has central angle 80° and radius 9.

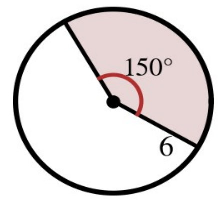
a) Find the sector's area exactly, then to the nearest tenth.



Example 3 – A Bigger Slice

In the middle figure, the sector has central angle 150° and radius 6.

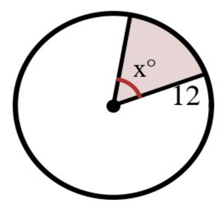
a) Find its area.



Example 4 – Working Backward

In the bottom figure, a sector of a radius-12 circle has area 24π.

a) Find x.



Example 5 – Pizza Math

A 14-inch pizza (that's the diameter) is cut into 8 equal slices.

a) Find the area of one slice to the nearest tenth.

b) What central angle does each slice have?

Example 6 – Scaling Question

a) Find the area of a circle with radius 4.

b) The radius doubles from 4 to 8. What happens to the area? The circumference?